EPS Rev B — integrated development schematic

2026-10-24: BMS pre-RBF block is now linked; ALL earlier BMS-absence statements are obsolete.

Checkpoint: controller, both converters, solar and rail protection. Physical RBF, charger PowerPath and BMS remain absent.

2026-10-24: PRE-CHARGER ABSENCE / EXTERNAL-SYS NOTES ABOVE AND BELOW ARE OBSOLETE.

Actual charger PowerPath now joins VIN_CHG and BATT_PERMITTED to SYS. Physical BMS/RBF hardware remains absent.

Controller

File: controller.kicad_sch

CAN interfaces

File: can_interfaces.kicad_sch

Essential 3V3

File: essential_3v3.kicad_sch

Charger PowerPath

File: charger.kicad_sch

Recovery 5V converter

File: recovery_5v.kicad_sch

Reference ranges: 100 controller, 200 BMS, 300 charger, 400 essential 3V3, 500 recovery 5V, 600 protection.

Solar source protection

File: source_protection.kicad_sch

Recovery rail protection

File: rail_protection.kicad_sch

2S pack protection

File: bms_protection.kicad_sch

SOLAR_PERMITTED -> source barrier -> VIN_CHG for FUTURE charger. VIN_CHG is NOT joined to SYS_PWR_PERMITTED.

SYS_PWR_PERMITTED remains an independent external permitted-source fixture. REG -> rail barrier -> protected+5V.

Incomplete ERC / low-VCC accuracy / hot SOA / shared startup /3V3 OV remain open. S03/S08/S11/S04 API limits; no PCB sync.

CURRENT: VIN_CHG -> Q300 -> R300 -> SYS; BATT_PERMITTED fixture -> Q301 -> SYS. No direct source bypass.

PROTECTED_PACK ends BEFORE missing RBF. BATT_PERMITTED remains a distinct external interlocked-source fixture.
Raw/system returns join only through BMS shunt; shared-positive SMBus supply leaves current-measurement closure open.

Sheet: /
File: eps_rev_b.kicad_sch

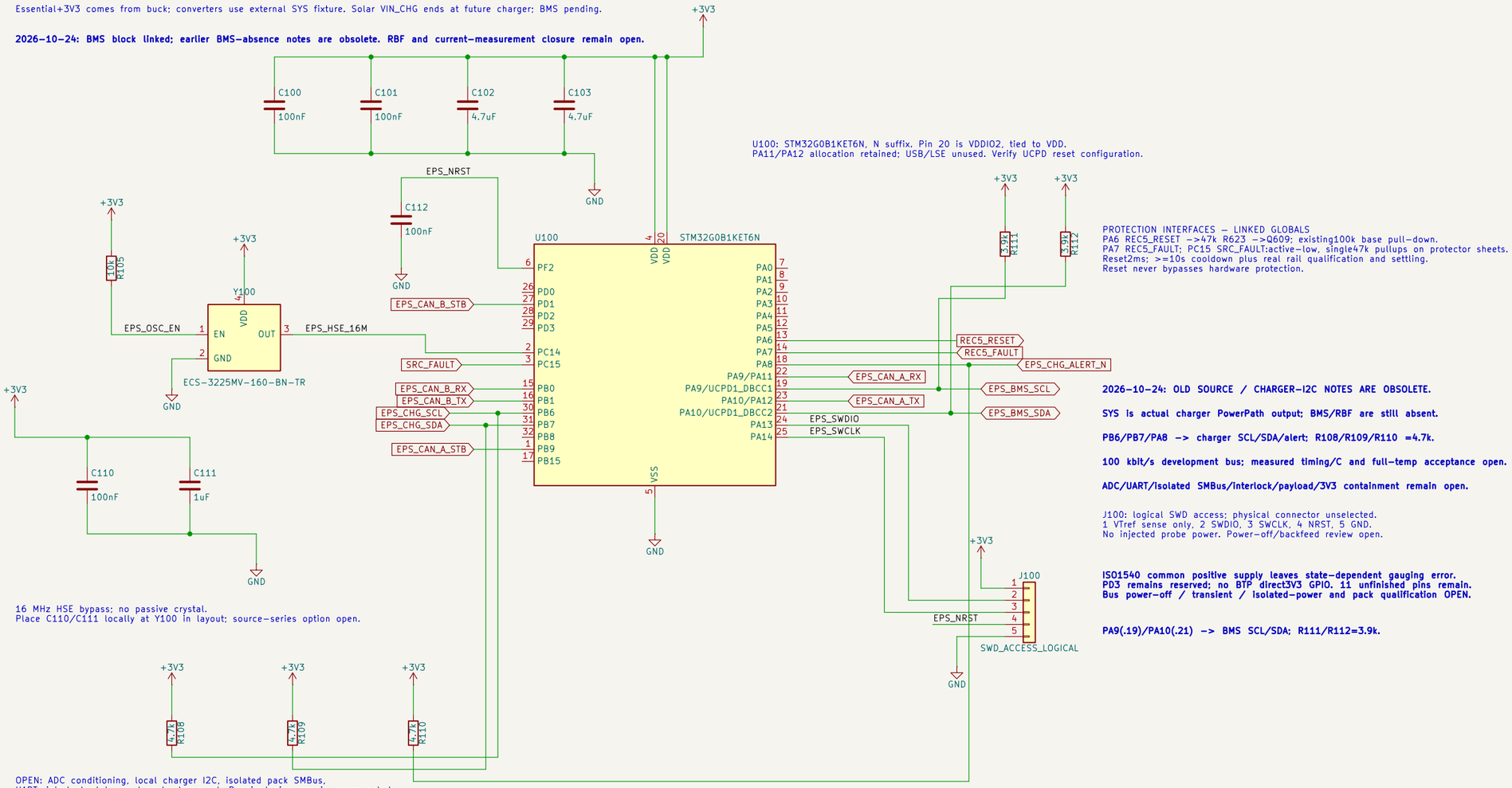
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KiCad E.D.A. 10.0.0		Id: 9/9

CONTROLLER — integrated development checkpoint

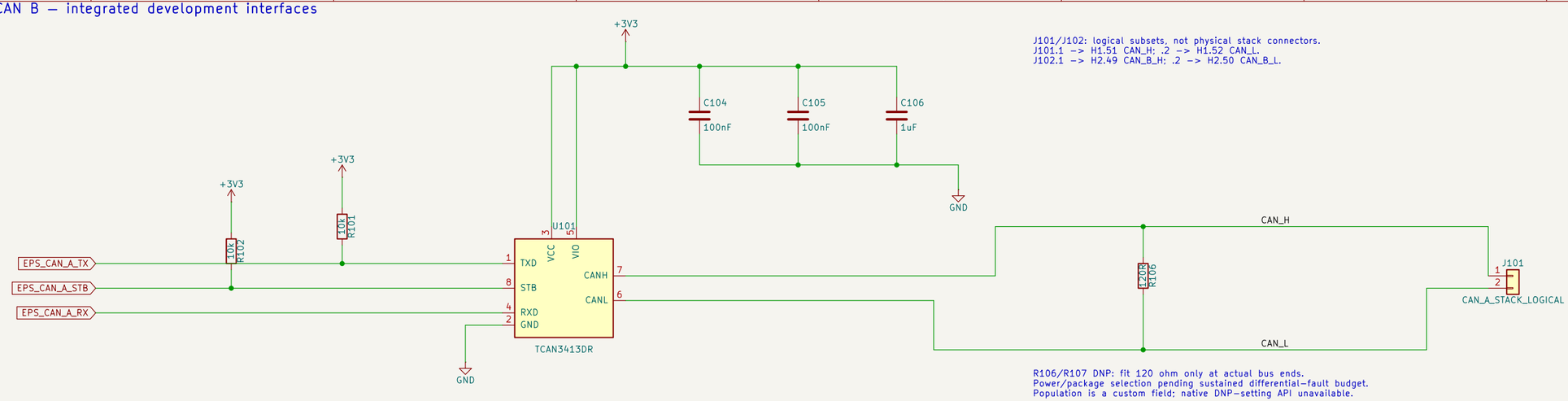
Essential+3V3 comes from buck; converters use external SYS fixture. Solar VIN_CHG ends at future charger; BMS pending.

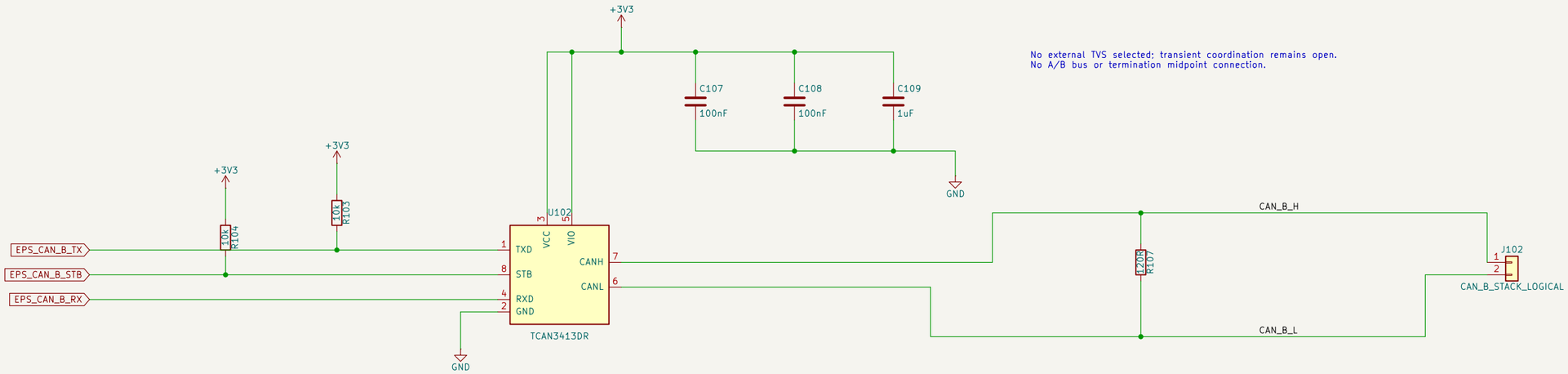
2026-10-24: BMS block linked; earlier BMS-absence notes are obsolete. RBF and current-measurement closure remain open.



Sheet: /Controller/		
File: controller.kicad_sch		
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Size: B	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 2/9

CAN A / CAN B — integrated development interfaces





FAULT LIMIT: shared +3V3, GND, MCU and clock remain common paths.
Independent VCC/VIO + pullup branch limiting remains a written candidate.
CAN B is bus-path recovery, not independent controller/power redundancy.

Existing TCAN custom symbols and manufacturer patterns reused.
Pin maps verified; assembly, ESD, timing and physical placement not qualified.
S08 construction review / S11 field-display limitations remain open.

Sheet: /CAN interfaces/		
File: can_interfaces.kicad_sch		
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Size: B	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 3/9

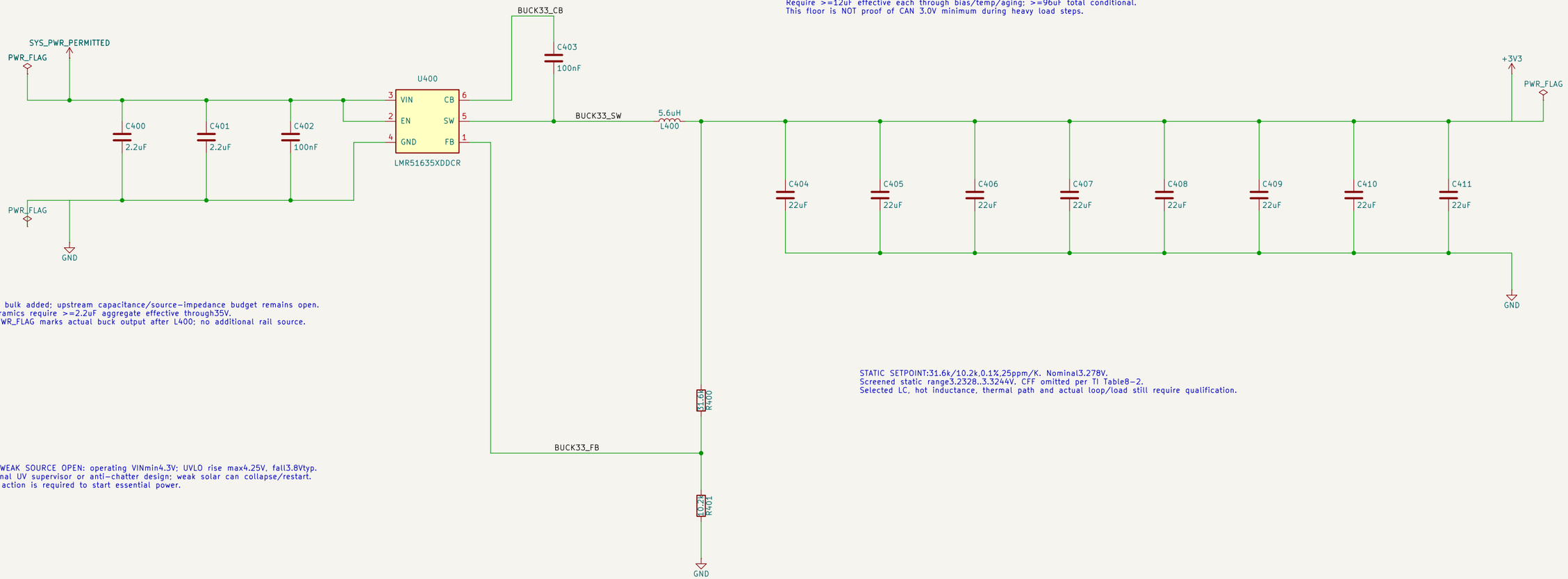
ESSENTIAL 3V3 — LMR51635XDDCR INTEGRATED DEVELOPMENT

EXTERNAL FIXTURE: SYS_PWR_PERMITTED and GND represent a bounded permitted source.
Input/return PWR_FLAGs do not implement source isolation, RBF, charger or battery protection.

5..35 V development envelope; 4.5 V sensitivity. EN = VIN; starts without MCU.
2 A is steady sizing only. Actual MCU/IHU/CAN ramp/load-step budget and transient acceptance remain open.

2026-10-24: EXTERNAL-SYS FIXTURE NOTE IS OBSOLETE: SYS now comes through real charger PowerPath.

#FLG0400 models that passive output. GND fixture remains; actual BMS/RBF and startup/source bounds remain open.



No input bulk added; upstream capacitance/source-impedance budget remains open.
Input ceramics require >=2.2uF aggregate effective through35V.
Output PWR_FLAG marks actual buck output after L400; no additional rail source.

OUTPUT: 8 x 22uF / 25V X7R, 176uF nominal.
Require >=12uF effective each through bias/temp/aging; >=96uF total conditional.
This floor is NOT proof of CAN 3.0V minimum during heavy load steps.

STATIC SETPOINT:31.6k/10.2k,0.1%,25ppm/K. Nominal3.278V.
Screened static range3.2328..3.3244V. CFF omitted per TI Table8-2.
Selected LC, hot inductance, thermal path and actual loop/load still require qualification.

UVLO / WEAK SOURCE OPEN: operating VINmin4.3V; UVLO rise max4.25V, fall3.8Vtyp.
No external UV supervisor or anti-chatter design; weak solar can collapse/restart.
No MCU action is required to start essential power.

OPEN AUTONOMOUS 3V3 OV CONTAINMENT: failed-short high-side can expose MCU/CAN
and25V output bank to-35V. Upstream solar OV protection does not contain this fault.
This stage does not establish complete EPS essential-power acceptance.

POLICY OPEN: S03 native severity and S11 field-display APIs unavailable.
Existing custom U400 reused; S08 original review-marker coverage unresolved. No PCB synchronization.

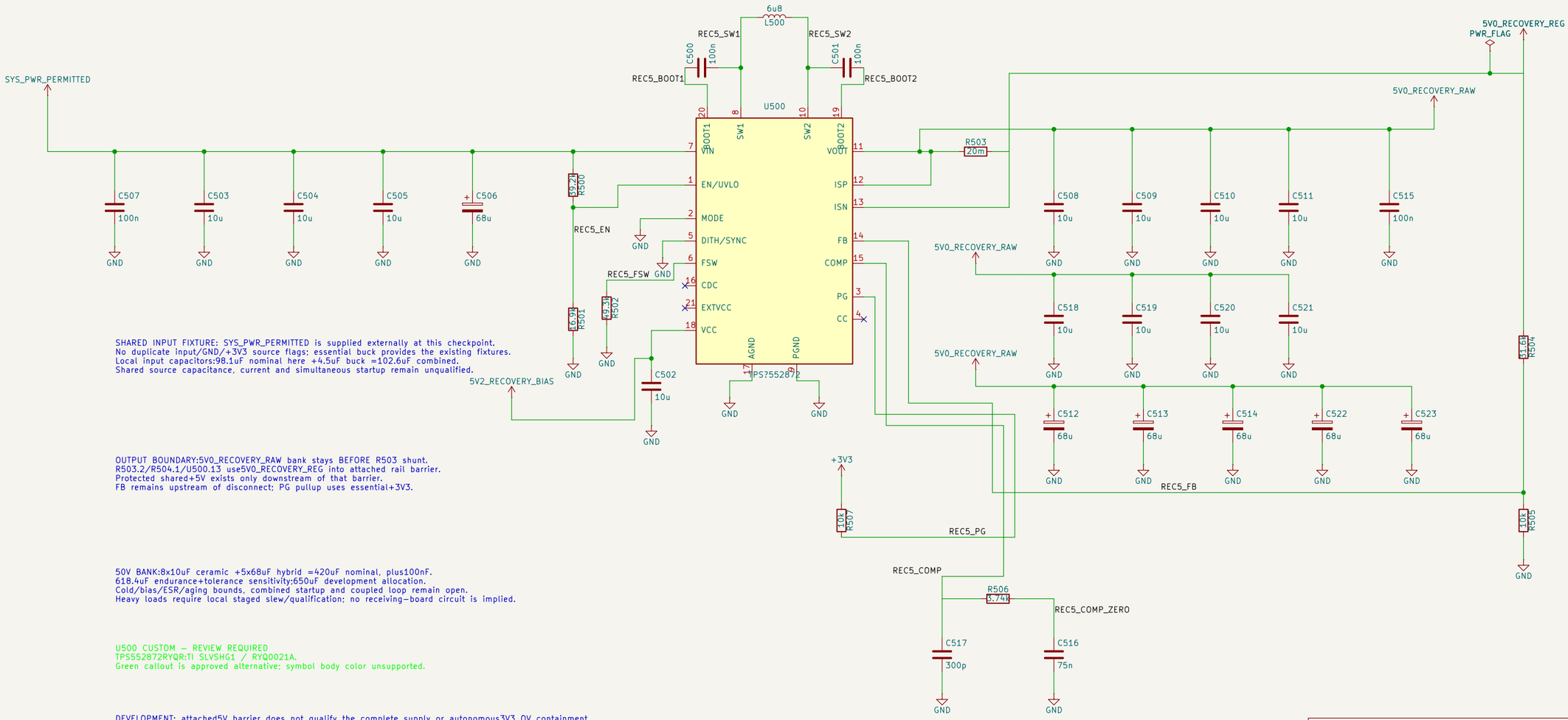
Sheet: /Essential 3V3/
File: essential_3v3.kicad_sch

Title:

Size: B | Date: | **Rev:**
KiCad E.D.A. 10.0.0 | Id: 4/9

RECOVERY 5V CONVERTER — INTEGRATED DEVELOPMENT

TPS552872RYQR: autonomous EN/UVLO, internal bias via floating EXTVCC; MODE low (PFM).
4.5..35 V development envelope: 2 A steady target only. Source/thermal/dynamic qualification remains open.



SHARED INPUT FIXTURE: SYS_PWR_PERMITTED is supplied externally at this checkpoint.
No duplicate input/GND/+3V3 source flags; essential buck provides the existing fixtures.
Local input capacitors:98.1uF nominal here +4.5uF buck =102.6uF combined.
Shared source capacitance, current and simultaneous startup remain unqualified.

OUTPUT BOUNDARY:5V0_RECOVERY_RAW bank stays BEFORE R503 shunt.
R503.2/R504.1/U500.13 use5V0_RECOVERY_REG into attached rail barrier.
Protected shared+5V exists only downstream of that barrier.
FB remains upstream of disconnect; PG pullup uses essential+3V3.

50V BANK:8x10uF ceramic +5x68uF hybrid =420uF nominal, plus100nF.
618.4uF endurance+tolerance sensitivity:650uF development allocation.
Cold/bias/ESR/aging bounds, combined startup and coupled loop remain open.
Heavy loads require local staged slew/qualification; no receiving-board circuit is implied.

U500 CUSTOM — REVIEW REQUIRED
TPS552872RYQR:TI SLVSHG1 / RYQ0021A.
Green callout is approved alternative; symbol body color unsupported.

DEVELOPMENT: attached5V barrier does not qualify the complete supply or autonomous3V3 OV containment.
SYS remains an external permitted fixture; physical RBF/charger/BMS absent. No PCB synchronization.

2026-10-24: PRIOR EXTERNAL-SYS / ABSENT-CHARGER NOTES ARE OBSOLETE. SYS now comes through charger PowerPath.

2026-10-24: BMS pre-RBF block linked; prior BMS-absence claim obsolete. RBF remains absent; battery source is still a fixture.

Sheet: /Recovery 5V converter/	
File: recovery_5v.kicad_sch	
Title:	
Size: B	Date:
KiCad E.D.A. 10.0.0	**Rev:** Id: 5/9

SOLAR SOURCE:33.5V analog clamp, timed cutoff, automatic retry

DEVELOPMENT ONLY. SOLAR_PERMITTED is an external interlocked-solar fixture; physical RBF is not drawn.

L600 limits input-short slew:require >=2.5uH through fault current.
C608=10nF timer. Bias/OV/snubber follow SRC_FILTERED.
Hot SOA, actual source energy and magnetic qualification remain open.

SOLAR_PERMITTED
PWR_FLAG
4.7u
L600
SRC_FILTERED
PWR_FLAG
Q600
IRLR2908
SRC_DG
SRC_COMMON
IRLR2908
Q601
SRC_SENSE
R600
20m
PWR_FLAG
VIN_CHG
SMBJ60A
D604
GND
SRC_GH
C600
68u
C601
10u
C602
10u
C603
10u
GND

SRC_FILTERED
R601
100
1N4148W-7-F
D600
PWR_FLAG
SRC_BIAS
C604
2.2u
C605
2.2u
C606
2.2u
C607
2.2u
GND

HGATE support / gate-source clamps / controlled startup

SRC_HGATE
SRC_COMMON
SRC_DG
SRC_SENSE
VIN_CHG
SRC_BIAS
U600
VCC
GATE
SOURCE
DGATE
SENSE
OUT
SHDN
FB
SRC_UV
SRC_OV
UV
OV
TMR
GND
ENOUT
FLT
LTC4364HDE-2
C608
10n
GND
VIN_CHG
25.8k
R606
SRC_FB
1k
R607
SRC_FAULT
+3V3
47k
R608
1N4148W-7-F
D601
SRC_HGATE
R609
4.7k
SRC_GH
MMBT3906LT1G
Q602
100
R610
SRC_COMMON
SRC_GH
SMBJ8.5A
D602
SRC_COMMON
SRC_DG
SMBJ8.5A
D603
SRC_COMMON
SRC_GH
C611
1n
2.2k
R611
GND

SRC_FAULT:active-low global to MCU PC15; single47k R608 pullup to essential+3V3.
SHDN and ENOUT intentionally NC; no MCU startup dependency.

SRC_BIAS
50k
R602
SRC_UV
10k
R603
C609
10n
GND

SRC_FILTERED
27.8k
R604
SRC_OV
1k
R605
C610
1n
GND

VIN_CHG is protected solar output for FUTURE charger PowerPath.
It is NOT joined to converter SYS_PWR_PERMITTED in this schematic.
Source output bank98uF nominal; downstream/source energy remains unqualified.

PV fixture <=1A; raw reservoir <=1uF:
harness<=1uH:60V abnormal test only.
SMBJ60A is not a blanket80V clamp.
Snubber>=2uF effective through60V;
actual source energy/hot leakage open.

SRC_FILTERED
2026-10-24: PRIOR FUTURE-CHARGER / SEPARATE-SYS NOTE IS OBSOLETE: VIN_CHG now feeds Q300/R300 PowerPath.

SYS is connected only through real PowerPath devices; source-loop/startup and BMS/RBF acceptance remain open.

R612
1.0
SRC_SNUB
C612
2.2u
C613
2.2u
C614
2.2u
C615
2.2u
GND

S03 native severity /S11 field presentation APIs remain open. Existing Panasonic review candidate reused.
No hot-SOA, arbitrary-source, physical-fit or complete EPS power qualification.

Sheet: /Solar source protection/
File: source_protection.kicad_sch
Title:
Size: B | Date: | **Rev:**
KiCad E.D.A. 10.0.0 | Id: 6/9

RECOVERY RAIL:5.270V analog clamp, timed latch-off

DEVELOPMENT: low-VCC FB accuracy, hot SOA and actual loads remain unqualified.

5V0_RECOVERY_REG
REC5_PIN
PWR_FLAG
4.7u
L601
Q603 IRLR2908
Q604 IRLR2908
Q605 IRLR2908
REC5_GH
REC5_DG
IRLR2908 Q606
REC5_COMMON
IRLR2908 Q607
IRLR2908 Q608
REC5_SENSE
R613 20m
+5V
PWR_FLAG
C616 100u
C617 100u
C618 22u
C619 22u
C620 22u
C621 22u
C622 22u
C623 22u
C624 22u
C625 22u
R625 330
GND

HGATE support / gate-source clamps / controlled startup

REC5_PIN
C627 2.2u
C628 2.2u
C629 2.2u
C630 2.2u
SMBJ36A D608

Three FETs per function reduce loss; equal linear-fault sharing is not assumed.

REC5_HGATE
REC5_COMMON
REC5_DG
REC5_SENSE
+5V
REC5_BIAS
U601
VCC
GATE
SOURCE
DGATE
SENSE
OUT
SHDN
FB
UV
OV
TMR
GND
ENOUT
FLT
LTC4364HDE-1
R617 3.57k
REC5_FB
R619 47k
+3V3
REC5_UV
R618 1.11k
REC5_FAULT
C632 100n

REC5_FAULT:active-low global to MCU PA7; single47k R619 pullup to essential+3V3.
SHDN/ENOUT intentionally NC. Reset PA6 acts through47k R623 and Q609.

1N4148W-7-F D605
REC5_HGATE
R620 4.7k
REC5_GH
MMBT3906LT1G Q610
R621 100
REC5_COMMON

SMBJ8.5A D606
REC5_GH
REC5_COMMON
SMBJ8.5A D607
REC5_DG
REC5_COMMON

REC5_PIN
R614 100
PWR_FLAG
REC5_BIAS
C631 100n
REC5_BIAS
R615 20k
REC5_UV
R616 10k
Q609 MMBT3904LT1G
REC5_RESET
R623 47k
R624 100k

Protected bank:2x100uF hybrid+8x22uF ceramic nominal.
Require>=112uF bulk /<=0.4ohm cold and>=48uF low-ESR ceramic;external<=100uF.
Actual cap/magnetic/pulse/source bounds and consumer voltage-time limits remain open.

5V0_RECOVERY_REG is actual converter output; converter FB and RAW bank stay UPSTREAM.
R625=330ohm bleed; C632=100nF timer; C626=10nF gate slew; R617/R618=3.57k/1.11k.
Require5ms heavy-load admission plus20ms settling and local rail qualification.
2ms UV reset never bypasses cooldown:>=10s development wait; restore<=0.5A startup state.

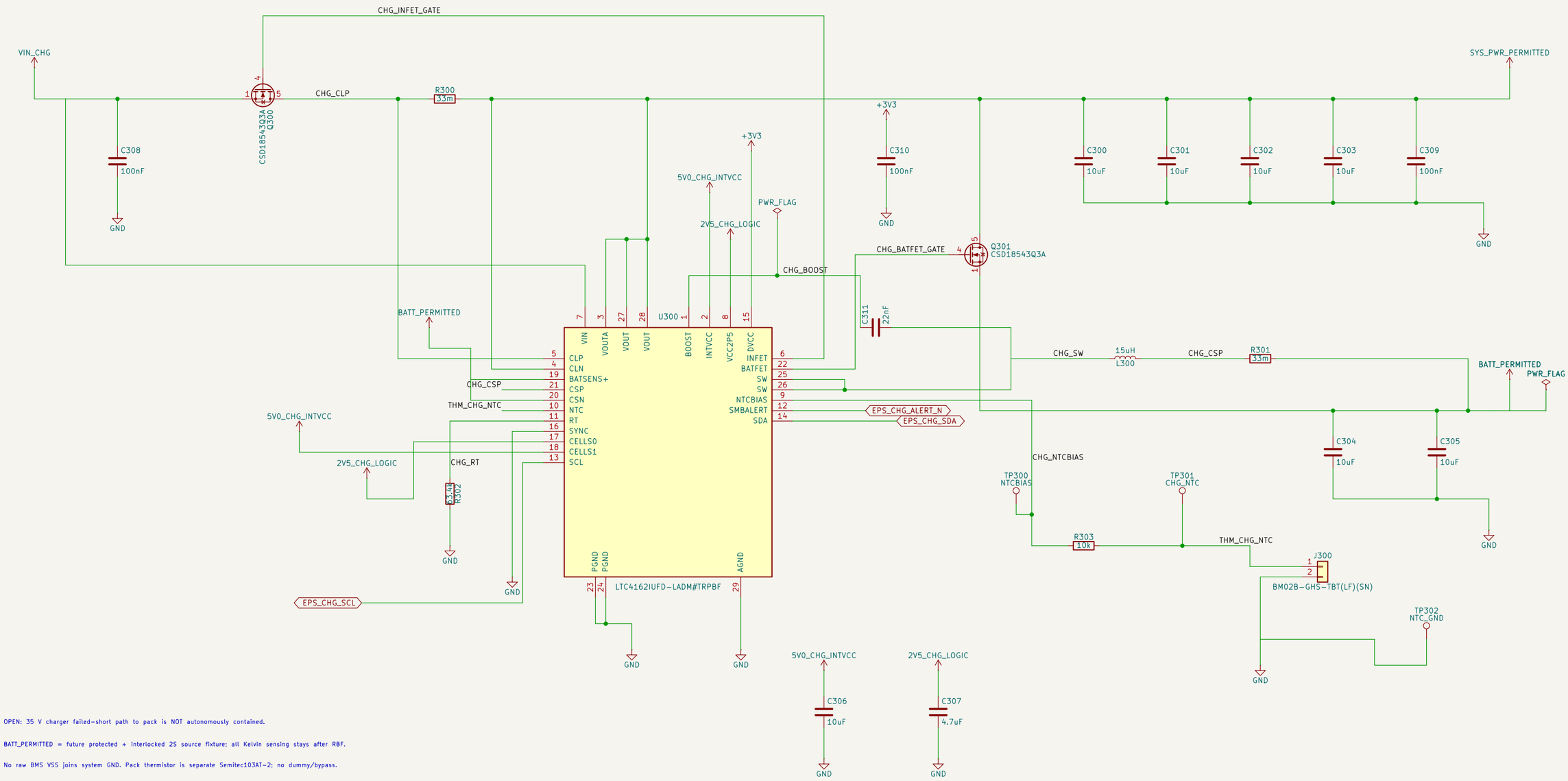
INTEGRATION IS NOT QUALIFICATION. Low-VCC clamp accuracy and hot linear SOA remain conditional.
Autonomous3V3 OV containment is absent. S03/S11 API limitations and reused-part review gates remain open.

C626 10n
R622 2.2k

Sheet: /Recovery rail protection/		
File: rail_protection.kicad_sch		
Title:		
Size: B	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 7/9

CUSTOM — REVIEW REQUIRED: LTC4162 symbol/land + both PowerPath footprints

DEVELOPMENT CHARGER: 2S straps; ~0.97 A charge setting; Input servo is NOT a source-current breaker



OPEN: 35 V charger failed-short path to pack is NOT autonomously contained.

BATT_PERMITTED = future protected + interlocked 2S source fixture; all Kelvin sensing stays after RBF.

No raw BMS VSS joins system GND. Pack thermistor is separate Semitec103AT-2; no dummy/bypass.

Input MPPT/weak-solar loop, shared startup, minimum effective C/L, hot SOA and sensor limits remain open.

SYS flag now models real passive PowerPath output; BOOST flag = internal diode + SW bootstrap energy.

No autonomous 3V3 OV or final BMS/RBF hardware is implied. Custom-field S11 presentation remains API-limited.

Sheet: /Charger PowerPath/		
File: charger.kicad_sch		
Title:		
Size: B	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 8/9

CUSTOM — REVIEW REQUIRED: U200 symbol/land; Q200/Q201 approved custom lands

2S PACK PROTECTION — DEVELOPMENT / NOT COMMISSIONED

Retained raw-cell domain: source is external cells/simulator.
R200 is the ONLY intended metal join to system GND.
No USB/debug ground across shunt; use differential probes.
Main-current connector/fuse and pack construction remain TBD.

300R precharge: <=28.9mA at8.526V illustrative bound.
8.526V is a sizing case, not an accepted pack OV limit.
No bypass around DSG or future RBF. Hot/pulse/SOA open.

PROTECTED_PACK is BEFORE future physical RBF.
BATT_PERMITTED remains separate; NO jumper.
35V charger-to-pack fault is NOT contained.
BQ configuration/pack thresholds uncommissioned.

BUS SUPPLY PROVISIONAL
Shared+3V3 positive;
distinct ground returns.
Raw bus current bypasses
shunt measurement: up to
8.42mA state-dependent.
Gauging NOT accepted.
Isolated power unresolved.
No whole-board galvanic
power-isolation claim.

Q203: low-Vth ground-gate development choice.
Reverse hotplug / cold gate / hot SOA remain open.

DSG short VGS can exceed20V; clamp / survival NOT accepted.

PBI needs >=2.2uF EFFECTIVE; C203 is conditional on bias/temp.
BAT reservoir hold-up / diode hot leakage remain open.
CG = Q203 reverse-charger clamp gate; ground bias1k.

TI SLUSCS4C Fig9-13: common-drain CHG/DSG arrangement.
2S: VC4/3/2 short AFTER top100R; VC1 reads midpoint.
SRP raw side / SRN system side; C202 is differential filtering.
No secondary fuse or PTC selected; permanent-failure policy TBD.
TS3/4, PRES/DISP/PTC/PTCEN and display disabled in configuration.

R200=2mOhm1%50ppm/K1WP70; Kelvin termination/pulse/cold open.
10M pulls retain TI driver-test load; hot leakage/turn-off open.
FUSE/BTP tie VSS perTI; output ERC retained, no type weakening.
Flag diamonds model actual raw sources/passive/internal paths.

100kbit/s: side1<=40pF, side2<=200pF.
Both sides >=3V +1ms before polling.
Reserve13.8mA IC +3.431mA pullups.
Static margins only; no rail-step proof.
Offset can reverse idle-current sign.

J200 SENSE ONLY:1RAW- 2MID 3RAW+ 4TS1 5RAW- 6TS2 7RAW-.
Two independent103AT-2 NTCs; quiet raw returns, no MCU bypass.
TS1/TS2 cell mode; TS2 default FET mode must change; commissioning TBD.

S03/S11 API limits remain. Ratings saved in metadata/BOM; review marker retained. Not fabrication/flight qualification.

C205 100nF
C206 100nF
BMS_ESD_FET_MID
TP200 TP_BATT_RAW
R207 5.1k
BMS_CHG_GATE
R209 10M
BMS_BAT_FILTERED
BATT_RAW
PWR_FLAG
Q200 CSD18543Q3A
R213 300
BMS_PCHG_LIMIT
Q202 BSS84-7-F
BMS_FET_DRAIN
PWR_FLAG
Q201 CSD18543Q3A
PWR_FLAG
PROTECTED_PACK
TP201 TP_PROTECTED_PACK
BATT_RAW
D200 BAT46W-7-F
C204 100nF
GND_BMS_RAW
R212 10M
BMS_PCHG_GATE
BMS_DSG_GATE
R210 10M
R208 5.1k
C207 100nF
BMS_ESD_PACK_MID
C208 100nF
GND
BMS_FET_DRAIN
PWR_FLAG
R205 100
BMS_VCC_FILTERED
PWR_FLAG
BMS_PBI
PWR_FLAG
R214 1k
CG
Q203 DMN62D2U-7
GND
BATT_RAW
R201 100
BMS_CELL_TOP_SENSE
PWR_FLAG
CELL1_RAW
TP204 TP_CELL1_RAW
C200 100nF
R202 100
BMS_CELL_MID_SENSE
C203 4.7uF
GND_BMS_RAW
R211 5.1k
C201 100nF
GND_BMS_RAW
U200 BQ40Z50RSMR-R2
BAT 32
VCC 26
PBI 1
VC4 2
VC3 3
VC2 4
VC1 5
SRP 8
SRN 6
TS1 10
TS2 11
TS3 12
TS4 13
PACK 27
PTC 23
PTCEN 24
PRES/SHUTDN 16
DISP 17
NC 7
NC 14
NC 29
VSS 9
VSS_EP 33
CHG 31
DSG 28
PCHG 30
FUSE 25
BTP_INT 15
SMBD 18
SMBC 19
LEDCNTLA 20
LEDCNTLB 21
LEDCNTLC 22
BMS_CHG_DRIVE
BMS_DSG_DRIVE
BMS_PCHG_DRIVE
BMS_SMBD
BMS_SMBC
BMS_LED_A
BMS_LED_B
BMS_LED_C
BMS_SRP
BMS_SRN
THM_BMS_TS1
THM_BMS_TS2
PROTECTED_PACK
R206 10K
BMS_PACK_SENSE
GND_BMS_RAW
R217 20k
R216 20k
R215 20k
GND_BMS_RAW
+3V3
C209 100nF
GND
+3V3
C210 100nF
GND_BMS_RAW
+3V3
R219 5.9k
+3V3
R218 5.9k
U201 ISO1540DR
VCC1 1
SDA1 2
SCL1 3
GND1 4
VCC2 8
SDA2 7
SCL2 6
GND2 5
EPS_BMS_SDA
EPS_BMS_SCL
BMS_SMBD
BMS_SMBC
GND
GND_BMS_RAW
C202 100nF
BMS_SRP
BMS_SRN
R203 100
R204 100
TP202 TP_GND_BMS_RAW
PWR_FLAG
R200 2m
TP203 TP_GND
GND_BMS_RAW
GND
BM07B-GHS-TBT(LF)(SN)
J200
7
6 THM_BMS_TS2
5
4 THM_BMS_TS1
3
2
1
CELL1_RAW
BATT_RAW
GND_BMS_RAW

Sheet: /2S pack protection/
File: bms_protection.kicad_sch
Title:
Size: B
Date:
Rev:
KiCad E.D.A. 10.0.0
Id: 9/9